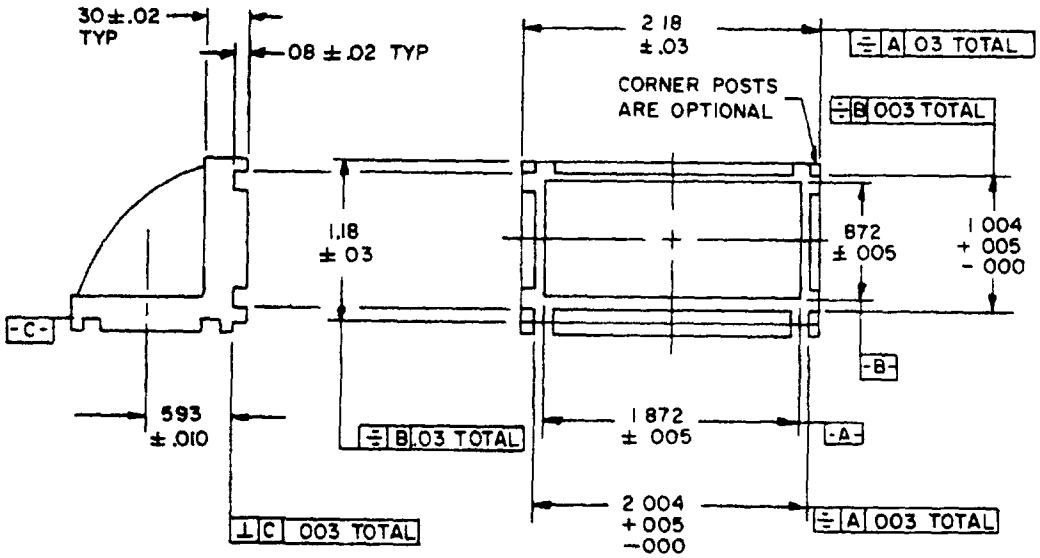
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, CAST, 90-DEGREE E-PLANE BEND, MITER AND RADIUS-BACK MITER, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

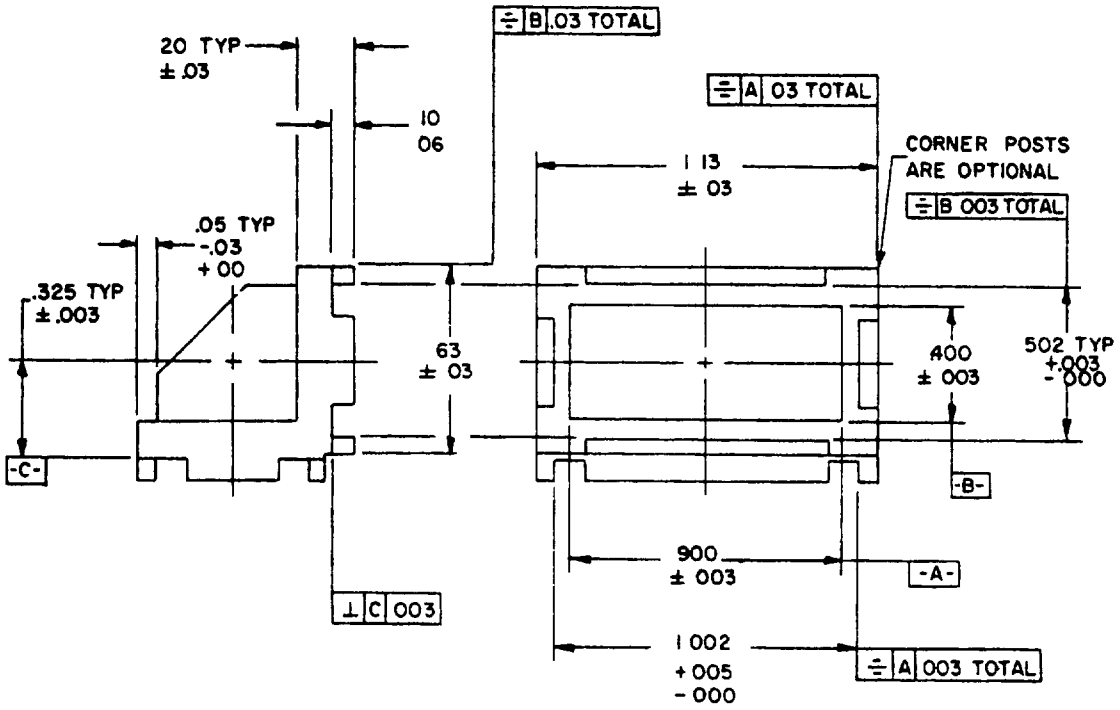


INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.08	2.0	1.18	30.0
.005	.13	.30	7.6	1.872	47.55
.010	.25	.593	15.06	2.004	50 90
.02	.5	.872	22 15	2.18	55.4
.03	.8	1.004	25.50		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Dimensions for radius-back miter configuration, for dash numbers 001 and 002.

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INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.06	1.5	.502	12.75
.005	.13	10	2.5	.63	16.0
.030	.76	.20	5.1	900	22.9
.03	.8	.325	8.26	1.002	25.45
.05	1.3	.400	10.16	1 13	28 7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Dimensions for miter configuration, for dash number 003

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TABLE I. Dash numbers and electrical and physical characteristics.

Dash no.	Frequency range TE_{10} mode GHz	VSWR max	Peak power kW max	Average power kW max	Operating temperature range	Pressurization lb_f/in^2 gage	Figure	Material	Mating waveguide M85/
001	4.0-5.9	1.05	1/	1/	0° to +60°C	20	1	Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371	1-053 1-054 1-174
002	4.0-5.9	1.05	1/	1/	0° to +60°C	20	1	Beryllium copper, alloy X6 per QQ-A-390	1-051 1-055
003	8.2-12.4	1.05	250	10	-55° to +200°C	45	2	Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371	1-077 1-078 1-178

1/ Assemblies shall have the same power handling capabilities as their mating waveguides.

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REQUIREMENTS:

Physical characteristics.
Dimensions and configuration· See figures 1 and 2.
Operating temperature range: See table I.
Material· See table I.
Surfaces (sandblasting after machining permissible):
Internal· 63√ microinches per ANSI B46.1.
External. 125√ microinches per ANSI B46.1.

Fungus resistance Materials nutrient to fungus shall not be used.

Pressurization See table I.

Penetrant inspection. No indication of any cracks at any location when tested per MIL-I-6866, type I, method B.

Electrical characteristics
Frequency range See table I.
VSWR See table I.
Power
Peak - See table I.
Average - See table I.

Part number M3970/23- (dash number from table I).

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent
DLA - ES

(Project 5985-0881-3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-W-3970/23

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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S/N 0102-014-1802

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